Omega 133 (2025) 103251

Contents lists available at ScienceDirect

Omega

journal homepage: www.elsevier.com/locate/omega

Risk control strategies for inventory pledge financing on cross-border e-commerce platforms empowered by the digital economy☆

Aolin Leng [a], Maolin Sun [b,c], Jinzhao Shi [b,c,*,#]

[a] School of Management, Northwestern Polytechnical University, Xi'an, 710072, China
[b] School of Economics and Management, Chang'an University, Xi'an 710064, China
[c] Research Center on Green Engineering and Sustainable Development, Chang'an University, Xi'an 710064, China

ARTICLE INFO

Keywords:
Digital economy
Cross-border e-commerce platform
Inventory pledge financing
Risk control

ABSTRACT

With the development of cross-border e-commerce (CBEC), CBEC-based inventory pledge financing (CBEC-based IPF) has gradually emerged, where exporters can apply for financing from cross-border e-commerce platforms (CBECPs) with their stocks in overseas warehouses. Due to the intensification of information asymmetry caused by overseas pledges and the impact of exchange rate fluctuations on repayments and settlements, such businesses face increased credit and market risks. Fortunately, the digital economy can empower CBECPs to achieve "post-loan risk management", such as using big data technology to monitor exporters' post-loan business activities on the platform and reduce their "credit default risk", or utilizing online marketing technologies such as search engine optimization and homepage recommendations to promote the sales of exporters' pledged goods and lower their "market sales risk". We focus on risk control issues in the CBEC-based IPF under the empowerment of the digital economy. The pre-loan risk control measures consider the setting of the pledge rate, while the post-loan risk control measures towards credit and market risks consider, respectively, the platform's online credit supervision and product sales assistance for exporters in light of the digital economy. On the basis of considering exchange rate fluctuations, we use optimization method to build decision models of a CBECP under two scenarios, i.e., "only control pre-loan risk" and "joint control pre- and post-loan risk", and find that compared to the former, the latter is more profitable for the CBECP, indicating that the digital economy can empower the CBECP to achieve more comprehensive risk control and improve economic benefits. Furthermore, in the case of the CBECP paying equal risk control cost, its optimal choices for credit and market risk control measures are provided. The results show that when and only when the market risk control is cheaper enough and the budget is limited, the CBECP chooses market risk control; otherwise, it always implements credit risk control. We also extend the model to study the case when the CBECP pays equal risk control level in the post-loan risk control, and conduct numerical experiments to verify the above conclusions. Our findings strongly suggest CBECPs to adopt the "joint pre- and post-loan risk control" in the CBEC-based IPF business, and provide them strategies for choosing between post-loan credit and market risk control measures.

1. Introduction

Cross-border e-commerce (CBEC) has grown rapidly in parallel with the rapid development of the Internet and the acceleration of economic globalization. This is now becoming a significant factor in China's foreign trade [1]. Chinese e-commerce giants, such as Alibaba and JD, have established cross-border e-commerce platforms (CBECPs), and an increasing number of conventional international trade firms are transitioning to CBEC [2]. According to data from 100ec, China's CBEC

☆ This manuscript was processed by Associate Editor Prof. Benjamin Lev

* Corresponding author.
E-mail addresses: aolin-leng@nwpu.edu.cn (A. Leng), 2021123026@chd.edu.cn (M. Sun), jzshi@chd.edu.cn (J. Shi).

Jinzhao Shi *PhD* is an associate professor in the School of Economics and Management at Chang'an University, Xi'an, China. He received his Ph.D degrees from Xi'an Jiaotong University and the City University of Hong Kong in 2019. His research interests include supply chain finance, low-carbon supply chain management, and so on. He has published articles in *International Journal of Production Economics, Computers & Industrial Engineering, Journal of the Operational Research Society, Expert Systems with Applications*, and others.

https://doi.org/10.1016/j.omega.2024.103251
Received 20 December 2023; Accepted 2 December 2024
Available online 9 December 2024
0305-0483/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Abbreviations

Abbreviations	Explanations/definitions
CBEC	cross-border e-commerce
CBECP	cross-border e-commerce platform
IPF	inventory pledge financing
PR	pledge rate, refers to the ratio of the loan amount to the value of the pledge

market had reached RMB 16.85 trillion in 2023, up 7.32 % year-on-year from 2022, with exports accounting for 78.59 % of the total.[1] Apart from the encouragement provided by national policies, the increasing maturity of cross-border logistics is one of the primary factors contributing to the rapid growth of CBEC. The overseas warehouse logistics mode is preferred by many exporters and buyers because of its quick delivery times and simple return and exchange policies. However, this mode is exposed to a higher inventory cost because it relies on overseas stock. In addition, CBEC traders mostly have characteristics such as "pure trade" and "light assets", making it difficult to obtain loans from commercial banks through the collateral of fixed assets, which puts them under greater financial pressure [2].

There are already some CBECPs in the market that, by utilizing the supply chain finance model, offer new financing possibilities for CBEC traders [3]. For example, JD Worldwide provides liquid asset pledge financing such as inventory pledge financing and accounts receivable financing for suppliers on its platform.[2] In the context of cross-border exports, the CBEC-based inventory pledge financing (CBEC-based IPF) has also emerged, through which exporters can finance from CBECPs based on their stocks in overseas warehouses to revitalize their inventories and expedite their capital turnover efficiency.[3] However, CBECPs are typically exposed to greater risks when providing such services to exporters because of information asymmetry exacerbated by overseas pledges and uncertain repayments and settlements under exchange rate fluctuations.[4] These risks include the credit risk of the exporters (the likelihood of exporters' default) and the market risk of the pledged goods (the uncertain demand for the pledged goods in overseas markets) [4]. *Therefore, controlling these risks becomes the primary concern for CBECPs.*

The risk control measures for traditional inventory pledge financing rely mostly on the setting of pledge rate (PR, the ratio of the loan amount to the value of the pledge) in the pre-loan approval stage. *However, with the development of big data technology, CBECPs empowered by the digital economy can realize post-loan risk control of exporters in CBEC-based IPF business. For instance, they can monitor exporters' post-loan business activities on the platform by using big data technology to reduce their "credit default risk", or use search engine optimization, homepage recommendations, and other online marketing technologies to promote the sales of exporters' pledged goods and reduce their "market sales risk".* Therefore, we focus on the risk control issues of CBEC-based IPF under the empowerment of the digital economy: (1) the risk control measures in the pre-loan approval stage considering the PR setting, and the two types of risk control measures in the post-loan management stage, credit and market, considering the CBECPs' credit supervision and product sales assistance for the exporters in light of the digital economy; (2) we construct decision-making models under two types of scenarios, namely, "only control pre-loan risk" and "joint control pre- and post-loan risk", to examine the value of the digital economy empowerment for CBECPs in loan risk control; and (3) in the case where CBECPs need to pay equal risk control cost, we provide CBECPs the optimal selection strategies for credit and market risk control in the post-loan management.

2. Literature review

Risk control in supply chain finance is a major concern for both industry and academia [5]. As a core indicator of risk control in the pre-loan approval stage, scholars have extensively studied the problem of decision-making and optimization of the PR, and their research can be categorized into two aspects. On one hand, scholars use tools such as GARCH, CVaR, Monte Carlo simulation, and heuristic algorithms to predict the price risk of pledged goods due to price fluctuations, and then construct PR optimization models [5–9]. Based on a data-driven copula model, Zhi et al. [5] set the optimal PR by assessing firms' probability of default and found that setting multiple PRs for different pledges can lead to higher profits from inventory financing. He et al. [6] predicted the value-at-risk (VaR) of pledges under different risk windows and set the corresponding dynamic PRs by building an AR-GARCH-GED model. Zhi et al. [7] developed a data-driven GARCH-EVT-Copula-based model that helps financial service providers set PRs and interest rates to improve their risk management in volatile market environments.

On the other hand, scholars have explored the risk control of supply chain finance from the perspective of supply chain operations by using methods such as game theory [10–26]. Focusing on the asset-based financing mode, Buzacott and Zhang [10] explored the risk management of banks and the inventory management of borrowing firms from a supply chain perspective. Li et al. [11] explored equity firms' risk management in the presence of capital constraints, demand uncertainty, and default risk. By constructing a Stackelberg game model between an e-commerce platform (the financial service provider) and an online retailer (the borrowing firm), Chang et al. [12] found that the e-commerce platform's ability to collect information about online transactions can provide a means of risk control to achieve a win-win situation for both parties. Sokolinskiy et al. [13] investigated the inventory management of borrowing firms in the presence of risks such as financing failure and supply disruption, while Xie et al. [14] explored the propagation effect of credit risk of borrowing firms by constructing a dual-channel financing model (bank credit and trade credit) and found that dual-channel financing reduced the propagation effect of credit risk compared with single-channel financing.

Generally, *most studies focus on the pre-loan PR optimization problem of traditional IPF, and few studies deal with the risk control problem in the post-loan management stage. However, we focus on the risk control issues of CBEC-based IPF under the empowerment of the digital economy, and build decision models under two scenarios, i.e., "only control pre-loan risk" and "joint control pre- and post-loan risk". Furthermore, in the case of CBECP paying equal risk control cost, the optimal choices for credit and market risk control measures are provided. Our research findings can provide decision-making reference for financial service providers such as CBECP to participate in joint pre- and post-loan risk control of CBEC-based IPF, and provide decision-making basis for their choice of post-loan credit or market risk control under limited costs.*

3. Problem description and assumptions

The borrower and the financial service provider of the CBEC-based IPF business in the context of overseas warehouses are the exporter and the CBECP, and Fig. 1 presents the detailed operational process of the business, where steps ①, ②, and ③ refer to the pre-loan approval

[1] https://www.100ec.cn/detail–6638736.html

[2] https://baijiahao.baidu.com/s?id=1741955327551192800&wfr=spider&for=pc

[3] In recent years, China has speeded up the construction of overseas warehouses. For example, Zongteng company has built >1.6 million square meters of overseas warehouses (https://www.goodcang.com). The application prospects of IPF in the context of overseas warehouses have also been widely discussed (https://scs.toocle.com/detail-1350.html).

[4] https://xueshu.baidu.com/usercenter/paper/show?paperid=136n0cv05t780rp0wv470pa0j3758940&site=xueshu_se.

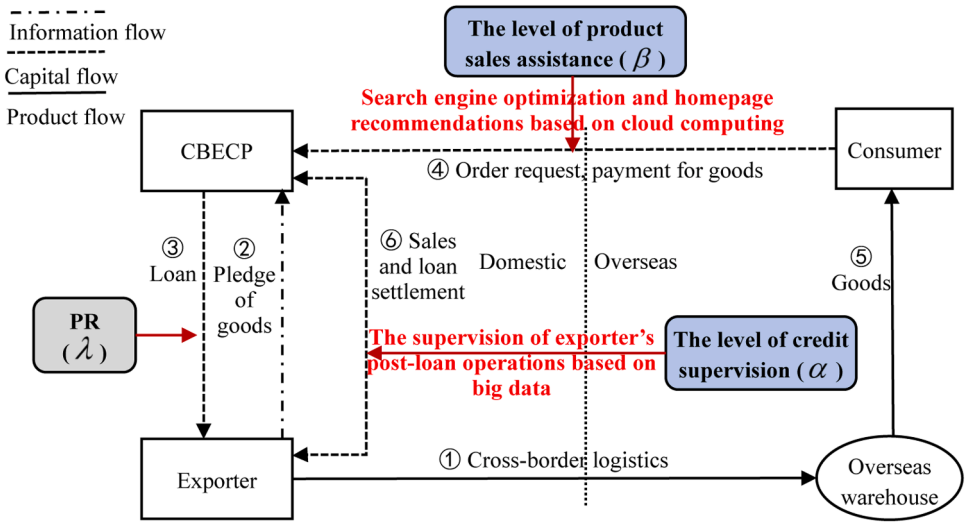


Fig. 1. The processes and risk control mechanisms of digital economy empowered CBEC-based IPF.

stage, while steps ④, ⑤, and ⑥ correspond to the post-loan management stage:

Step ①. The domestic exporter uses cross-border logistics to transport its goods to overseas warehouse, which is supervised by logistics enterprise designated by the CBECP.
Step ②. To alleviate liquidity shortages, the exporter pledges goods in overseas warehouse to the CBECP and apply for IPF.
Step ③. The CBECP determines the PR, taking into account the quantity and price of the pledge, exchange rate fluctuations, and the exporter's credit level, and informs the exporter of the PR. If the exporter agrees with the PR provided by the CBECP, both parties sign a contract, and then the CBECP issues a loan to the exporter.
Step ④. During the sales season, overseas consumers request orders from the CBECP and pay for the goods in their local currency.
Step ⑤. The CBECP instructs logistics enterprise to release goods from overseas warehouse and deliver them to overseas consumers.
Step ⑥. The CBECP settles the sales and loan with the exporter. If the sales of the pledge are greater than the principal and interest of the loan, the CBECP will deduct the principal and interest of the loan and the remainder will be the exporter's sales profit; if the sales of the pledge are less than the principal and interest of the loan, the exporter will choose to default on the loan with a certain probability.

Table 1
Summary of notations.

Notations	Definitions
General/global notations	
p	Retail price per unit of product (in domestic currency)
X	Stochastic market demand
x	Realized market demand, $x \in X$
$f(x), f'(x), F(x)$	Probability density function of X and its derivative, cumulative distribution function
Q	Pledged amount of product of the exporter
R	Loan interest rate provided by CBECP, $R > 0$
ξ	Default probability of the exporter, $0 \le \xi \le 1$
θ_0	The exchange rate during the sales season (1 unit of local currency for θ_0 units of foreign currency)
θ_1	The exchange rate at the moment of settlement between CBECP and the exporter (1 unit of local currency for θ_1 units of foreign currency)
θ	$\theta = \theta_1 / \theta_0$, measuring degree of the appreciation of domestic currency
Only control pre-loan risk scenario	
D	Market demand
$\hat{x}$	Demand threshold for exporter's loan default
λ	PR of CBECP, measuring the ratio of loan amount to the value of pledge
π	Expected profit of CBECP
Joint control pre- and post-loan risk scenario	
(1) Post-loan credit risk control	
D_c	Market demand
$\hat{x}_c$	Demand threshold for exporter's loan default
λ_c	PR of CBECP
α	The level of credit supervision of CBECP, $\alpha \ge 0$
k_c	Cost coefficient of credit supervision of CBECP
π_{CRC}	Expected profit of CBECP
(2) Post-loan market risk control	
D_m	Market demand
$\hat{x}_m$	Demand threshold for exporter's loan default
λ_m	PR of CBECP
β	The level of product sales assistance of CBECP, $\beta \ge 0$
k_m	Cost coefficient of product sales assistance of CBECP
π_{MRC}	Expected profit of CBECP

In the traditional offline IPF business, the financial service providers such as banks are outside the production and operations of the borrowing enterprises, which leads to an obvious information asymmetry between the two parties. Therefore, the financial service providers usually adopt only the method of PR to control the loan risk at the stage of "pre-loan approval", while lacking effective "post-loan risk control" measures [6]. However, in the CBEC-based IPF, the exporter's operations are closely related to the CBECP, and its logistics, information flow, capital flow and other information can be effectively captured to a certain extent by the CBECP. *Further digitization and modeling of such information can provide new opportunities for CBECP to realize "post-loan risk control". This means that in addition to the PR control at the "pre-loan approval" stage (step ③ in Fig. 1), the digital economy can empower CBECP to realize "post-loan risk control". For instance, CBECP can use big data technology to monitor the exporter's post-loan business activities and flow of funds on the platform and reduce exporter's "credit default risk" (step ⑥ in Fig. 1)* .[5] *Alternatively, search engine optimization, homepage recommendations, and other online marketing techniques can be used to promote the sales of exporter's pledged goods and reduce its "market sales risk" (step ④ in Fig. 1).*

To better capture the value of the digital economy in enabling CBECP to realize "post-loan risk control", we consider the following two scenarios: *First, in the "only control pre-loan risk" scenario* (Section 4), the

[5] Note that the monitoring of credit risk runs through all post-loan stages, but it is ultimately reflected in the default probability of the exporter in step ⑥.

CBECP chooses to carry out only the pre-loan risk control by setting the PR, i.e., λ; *second, in the "joint control pre- and post-loan risk" scenario* (Section 5), the CBECP adopts both pre-loan risk control (setting the PR of λ) and post-loan risk control (setting credit supervision level α in the credit risk control or product sales assistance level β in the market risk control), and pays the corresponding costs for different post-loan risk control measures.

By comparing the CBECP's returns in the above two scenarios, we mainly answer the two key questions in Section 6: (1) Is "joint control pre- and post-loan risk" more beneficial to the CBECP, compared to "only control pre-loan risk"? In other words, can joint risk control empowered by the digital economy improve the benefits of the CBECP? (2) Which alternative post-loan risk control measure, i.e., credit or market, performs better so that the CBECP should choose in practice?

To facilitate the study, we made the following assumptions.

A1. There is a period between the end of the sales season and settlement between the CBECP and the exporter, during which the exchange rates between the two countries may change [27]. The exchange rate is predictable and remains unchanged throughout the sales season.
A2. If the market demand is greater than the demand threshold for the exporter's default, the exporter will not default at the end of the selling season. Conversely, it chooses to default with the probability of ξ. The exporter will only undertake "limited liability" in case of default [28,29].
A3. The pledged goods are seasonal products, and their residual values and stockout costs are not considered [30,31].
A4. The profits of CBECP are measured in domestic currency [27].
A5. The probability density function of the random market demand is non-increasing, which means that $f'(x) \leq 0$.

Table 1 summarizes the notations.

4. Only control pre-loan risk scenario

When the CBECP only carries out pre-loan risk control, it only decides on setting the PR at the pre-loan approval stage and does not take any risk control measures at the post-loan management stage ($\alpha = 0$and $\beta = 0$). Under a given PR of λ, the CBECP provides loans to an exporter in the amount of λpQ. At the end of the loan period, the exporter has to pay the principal and interest of the loan, i.e., $\lambda pQ(1+R)$, to the CBECP. The market demand in this case is represented by $D = X$. Based on Assumption A1, when the CBECP settles with the exporter after sales, the actual payment that the exporter receives can be expressed in the domestic currency as $\frac{p\theta_0}{\theta_1}\min(D, Q)$. Thus, the demand threshold for exporter default is $\hat{x} = \frac{\lambda Q(1+R)\theta_1}{\theta_0}$, below which the exporter will choose to default with a probability of ξ, according to Assumption A2. The profit function of the CBECP can then be expressed as

$$\begin{aligned}\pi(\lambda) &= \underbrace{\xi\int_0^{\hat{x}}\frac{px\theta_0}{\theta_1}f(x)dx + (1-\xi)\int_0^{\hat{x}}\frac{p\hat{x}\theta_0}{\theta_1}f(x)dx}_{(i)} \\ &\quad + \underbrace{\int_{\hat{x}}^{+\infty}\frac{p\hat{x}\theta_0}{\theta_1}f(x)dx}_{(ii)} - \underbrace{\lambda pQ}_{(iii)} \\ &= \lambda pQR - \frac{\xi p}{\theta}\int_0^{\hat{x}}F(x)dx\end{aligned} \quad . \tag{1}$$

In Eq. (1), the term (*i*) is the expected revenue of the CBECP when the actual market demand is less than $\hat{x}$, which contains both cases in which the exporter chooses to default or fulfill its obligations under a certain probability; the term (*ii*) is the expected revenue of the CBECP when the actual market demand is greater than $\hat{x}$; while the term (*iii*) is the cost of loan to the CBECP. The maximum value of Eq. (1) leads to Proposition 1.

Proposition 1. *When the CBECP only performs pre-loan risk control, its optimal PR is*

$$\lambda^* = \frac{F^{-1}(\delta)}{Q(1+R)\theta}, \tag{2}$$

where $\delta = \frac{R}{\xi(1+R)}$.

Proposition 1 provides the optimal PR for the CBECP under only control pre-loan risk scenario. It can be seen that the CBECP's optimal PR is determined by the loan interest rate R, the pledged amount of the product Q, the default probability of the exporter ξ, the degree of appreciation of domestic currency θ, and the distribution of the random market demand $f(x)$. It is surprising to find that its optimal PR is independent of the selling price of the product p, which differs from the classical newsvendor problem. Based on Proposition 1, we further have $\frac{d\lambda^*}{dQ} < 0$, $\frac{d\lambda^*}{d\xi} < 0$, and $\frac{d\lambda^*}{d\theta} < 0$, which suggest that the CBECP will lower its PR to more strictly control the loan risk in the case of a higher pledged amount, a higher default probability of the exporter, or a higher degree of appreciation of the domestic currency.

5. Joint control pre- and post-loan risk scenario

Empowered by the digital economy, the CBECP can conduct joint pre- and post-loan risk control. In this case, the CBECP determines the PR at the pre-loan approval stage and implements different risk control measures at the post-loan management stage for credit and market risks, such as setting the level of credit supervision for credit risk control or the level of product sales assistance for market risk control.

5.1. "Post-loan credit risk" control

In the case of post-loan credit risk control, the CBECP will reduce the exporter's probability of default by conducting credit supervision at the post-loan management stage, i.e., using big data technology to monitor the exporter's post-loan business activities on the platform. The cost of credit supervision generated for the CBECP is $C_{CRC}(\alpha) = \frac{1}{2}k_c\alpha^2$, where $C_{CRC}'(\alpha) > 0$ and $C_{CRC}''(\alpha) > 0$. The influencing factor of credit supervision on the default probability of the exporter in case of poor market is $\psi_{CRC}(\alpha) = \frac{1}{1+\alpha}$. The market demand in this scenario is still $D_c = X$, so the exporter is still expected to receive $\frac{p\theta_0}{\theta_1}\min(D_c, Q)$ when the sales are settled. Therefore, the demand threshold for exporter default in this case is $\hat{x}_c = \frac{\lambda_c Q(1+R)\theta_1}{\theta_0}$, below which the exporter will choose to default with a probability of $\psi_{CRC}(\alpha)\cdot\xi$. Furthermore, the profit function of the CBECP changes to

$$\begin{aligned}\pi_{CRC}(\lambda_c, \alpha) &= \underbrace{\psi_{MRC}(\alpha)\xi\int_0^{\hat{x}_c}\frac{px\theta_0}{\theta_1}f(x)dx + [1-\psi_{MRC}(\alpha)\xi]\int_0^{\hat{x}_c}\frac{p\hat{x}_c\theta_0}{\theta_1}f(x)dx}_{(i)} \\ &\quad + \underbrace{\int_{\hat{x}_c}^{+\infty}\frac{p\hat{x}_c\theta_0}{\theta_1}f(x)dx}_{(ii)} - \underbrace{[\lambda_c pQ + C_{CRC}(\alpha)]}_{(iii)} \\ &= \lambda_c pQR - \frac{\xi p}{(1+\alpha)\theta}\int_0^{\hat{x}_c}F(x)dx - \frac{1}{2}k_c\alpha^2\end{aligned} \quad . \tag{3}$$

In Eq. (3), the terms (*i*) and (*ii*) are the expected returns of the CBECP

in the cases of $x < \hat{x}_c$ and $x > \hat{x}_c$ respectively, and the term (*iii*) is the loan cost and the credit supervision cost of the CBECP. The maximum value of Eq. (3) leads to Proposition 2.

Proposition 2. *When the CBECP performs joint pre-loan and post-loan-credit risk control, its optimal PR and credit supervision level are*

$$\begin{cases} \lambda_c^* = \dfrac{F^{-1}[\delta(1+\alpha^*)]}{Q(1+R)\theta} \\ \alpha^* = \dfrac{\xi p \int_0^{F^{-1}[\delta(1+\alpha^*)]} F(x)dx}{\theta k_c (1+\alpha^*)^2} \end{cases}, \tag{4}$$

where $\delta = \frac{R}{\xi(1+R)}$.

Then, by comparing λ_c^* in Eq. (4) and λ^* in Eq. (2), we get the following proposition.

Proposition 3. *The CBECP's optimal PR in the joint pre-loan and post-loan-credit risk control is strictly larger than that in the only pre-loan risk control,* i.e., $\lambda_c^* > \lambda^*$.

Proposition 3 shows that under the joint pre-loan and post-loan-credit risk control, the digital economy can empower the CBECP to achieve post-loan credit risk management, which allows the CBECP to relax its control over the pre-loan PR, that is, to increase the PR. Of course, this will be advantageous for the exporter as a borrower, as it means he/she will receive more loans with the same number of pledges, or an equal loan with a smaller number of pledges. From analyzing the results in Eq. (4), we can also get the following result towards the relationship between λ_c^* and α^*.

Proposition 4. *When joint control pre-loan and post-loan-credit risk, the CBECP's PR is positively related to its credit supervision level,* i.e., $d\lambda_c^*/d\alpha^* > 0$.

Proposition 4 implies that in the joint risk control, the CBECP will choose to only strengthen one of the pre-loan and post-loan risk control measures, that is, when strengthening the pre-loan risk control, it will relax the post-loan risk control, or vice versa. For example, when α^* grows, the post-loan credit supervision is strengthened, which will bring about a synchronous rise in the PR λ_c^*, relaxing the pre-loan risk control.

5.2. "Post-loan market risk" control

In the case of post-loan market risk control, the CBECP is required to carry out product sales assistance, e.g., utilizing online marketing technologies such as search engine optimization and homepage recommendations, to increase demand in overseas markets and promote the sales of the exporter's pledged goods. The cost of product sales assistance that generated for the CBECP is $C_{MRC}(\beta) = \frac{1}{2}k_m\beta^2$, where $C_{MRC}'(\beta) > 0$ and $C_{MRC}''(\beta) > 0$. The influencing factor of product sales assistance on the market demand is $\psi_{MRC}(\beta) = 1+\beta$. The market demand in this scenario changes into $D_m = \psi_{MRC}(\beta)X$, so the exporter is expected to receive $\frac{p\theta_0}{\theta_1}\min(D_m, Q)$ when the sales are setttled. Therefore, the demand threshold for exporter default in this case is $\hat{x}_m = \frac{\lambda_m Q(1+R)\theta_1}{\psi_{MRC}(\beta)\theta_0}$, below which the exporter will choose to default with a probability of ξ. Further, the profit function of the CBECP is changed into

$$\begin{aligned} \pi_{MRC}(\lambda_m,\beta) &= \underbrace{\xi\int_0^{\hat{x}_m}\frac{p\psi_{MRC}(\beta)x\theta_0}{\theta_1}f(x)dx + (1-\xi)\int_0^{\hat{x}_m}\frac{p\psi_{MRC}(\beta)\hat{x}_m\theta_0}{\theta_1}f(x)dx}_{(i)} \\ &+ \underbrace{\int_{\hat{x}_m}^{+\infty}\frac{p\psi_{MRC}(\beta)\hat{x}_m\theta_0}{\theta_1}f(x)dx}_{(ii)} - \underbrace{[\lambda_m pQ + C_{MRC}(\beta)]}_{(iii)} \\ &= \lambda_m pQR - \frac{\xi p(1+\beta)}{\theta}\int_0^{\hat{x}_m}F(x)dx - \frac{1}{2}k_m\beta^2 \end{aligned}. \tag{5}$$

Similar to Eq. (3), in Eq. (5), the terms (*i*) and (*ii*) are the expected returns of the CBECP in the case of $x < \hat{x}_m$ and $x > \hat{x}_m$ respectively, and the term (*iii*) is the loan cost and the product sales assistance cost of the CBECP. The maximum value of Eq. (5) leads to Proposition 5.

Proposition 5. *When the CBECP performs joint pre-loan and post-loan-market risk control, its optimal PR and product sales assistance level are*

$$\begin{cases} \lambda_m^* = \dfrac{F^{-1}(\delta)(1+\beta^*)}{Q(1+R)\theta} \\ \beta^* = \dfrac{\xi p\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]}{\theta k_m} \end{cases}, \tag{6}$$

where $\delta = \frac{R}{\xi(1+R)}$.

Then, by comparing λ_m^* in Eq. (6) and λ^* in Eq. (2), we get the following proposition.

Proposition 6. *The CBECP's optimal PR in the joint pre-loan and post-loan-market risk control is strictly larger than that in the only pre-loan risk control,* i.e., $\lambda_m^* > \lambda^*$.

Like Proposition 3, Proposition 6 shows that under the joint pre-loan and post-loan-market risk control, the digital economy can also empower the CBECP to achieve post-loan market risk management, which allows the CBECP to relax its control over the pre-loan PR, that is, to increase the PR. From analyzing the results in Eq. (6), we get the following results.

Proposition 7. *When joint control pre-loan and post-loan-market risks,*

(a) the CBECP's PR is positively related to its product sales assistance level, i.e., $d\lambda_m^*/d\beta^* > 0$.
(b) the CBECP's PR and product sales assistance level are all negatively correlated with the exporter's probability of default, i.e., $d\lambda_m^*/d\xi < 0$ and $d\beta^*/d\xi < 0$.
(c) the CBECP's PR and product sales assistance level are all negatively correlated with the degree of appreciation of the domestic currency, i.e., $d\lambda_m^*/d\theta < 0$ and $d\beta^*/d\theta < 0$.

Proposition 7(a) also implies that in the joint risk control scenario, the CBECP will choose to strengthen only one of the pre- and post-loan risk control measures. For example, when β^* grows, the post-loan product sales assistance is strengthened, which will bring about a synchronous rise in the PR λ_m^*, relaxing the pre-loan risk control. Proposition 7(b) further illustrates that as the probability of the exporter default increases, the CBECP tends to strengthen the pre-loan risk control by reducing the PR; otherwise, it will tend to strengthen the post-loan risk control by improving the level of product sales assistance. Similarly, Proposition 7(c) suggests that as the degree of appreciation of the

domestic currency increases, the CBECP will be more inclined to strengthen the pre-loan risk control, rather than the post-loan risk control.

6. Strategy selection for post-loan risk control

In this section, we further answer two key questions. One is whether "joint control pre- and post-loan risk" is more beneficial for the CBECP compared to the "only control pre-loan risk", that is, whether digital economy empowered joint risk control can improve the overall efficiency of the CBECP? The second is which post-loan risk control measure is more advantageous for the CBECP when it needs to pay the same level of risk control cost?

By comparing the optimal profits of the CBECP under two scenarios, i.e., "joint control pre- and post-loan risk scenario" in Section 5 and "only control pre-loan risk scenario" in Section 4, we can get Proposition 8.

Proposition 8. *The "joint pre- and post-loan risk control" always brings additional benefits to the CBECP, whether implementing credit or market risk control strategies in the post-loan stage.*

Proposition 8 shows that the "joint pre- and post-loan risk control" is more beneficial for the CBECP, which also indicates that digital economy can empower the CBECP to achieve more comprehensive risk control in the CBEC-based IPF business and improve economic efficiency. Meanwhile, from the results in Propositions 3 and 6, the CBECP will provide a higher PR for the borrower in the joint risk control, meaning that the exporter can obtain more loan resources. Therefore, with the support of the digital economy, the CBECP can achieve a new breakthrough in the post-loan risk control, which benefits both the borrower and the lender.

When the CBECP pays the same cost for post-loan credit and market risk control, the optimal choice strategy for the CBECP to carry out post-loan credit or market risk control can be obtained by comparing the profits of the CBECP in the two types of scenarios. Thus, we get Proposition 9.

Proposition 9. *In the "joint pre- and post-loan risk control" scenario, when the CBECP pays the same cost for post-loan credit and market risk control (credit supervision and product sales assistance),* i.e., $C_{CRC}(\alpha) = C_{MRC}(\beta) = C$, *there exists a certain* $\omega = \frac{\int_0^{F^{-1}(\delta)} F(x)dx}{\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)} F(x)dx} > 1$,

(a) If $k_c \le \omega^2 k_m$, the CBECP should choose post-loan credit risk control.

(b) If $k_c > \omega^2 k_m$, there exists a certain threshold $\widetilde{\alpha} \in \left(0, \frac{1-\delta}{\delta}\right)$ that solves Eq. (7), and when $C > C_{CRC}(\widetilde{\alpha})$, the CBCEP should choose post-loan credit risk control; when $C < C_{CRC}(\widetilde{\alpha})$, the CBECP should choose post-loan market risk control.

$$\delta\left\{F^{-1}[\delta(1+\widetilde{\alpha})] - (1+\Omega\widetilde{\alpha})F^{-1}(\delta)\right\} = \frac{1}{(1+\widetilde{\alpha})}\int_0^{F^{-1}[\delta(1+\widetilde{\alpha})]} F(x)dx - (1+\Omega\widetilde{\alpha})\int_0^{F^{-1}(\delta)} F(x)dx. \tag{7}$$

Where $\delta = \frac{R}{\xi(1+R)}$ and $\Omega = \sqrt{\frac{k_c}{k_m}}$.

Proposition 9 shows that when the CBECP pays the same cost for credit and market risk control, its optimal choice depends on the cost coefficients of the credit supervision k_c and product sales assistance k_m, as well as the level of the risk control cost or budget. When the cost coefficient of credit supervision k_c is relatively small, i.e., $k_c \le \omega^2 k_m$, the CBECP chooses post-loan credit risk control. Otherwise, there exists a budget threshold $C_{CRC}(\widetilde{\alpha})$, when the CBECP's budget for post-loan risk control is above such a threshold, it still chooses credit risk control; otherwise, it chooses market risk control. *In summary, when and only when the market risk control is cheaper enough (cost coefficient of product sales assistance satisfies* $k_m < k_c/\omega^2$*) and budget for post-loan risk control is relatively limited (risk control cost meets* $C < C_{CRC}(\widetilde{\alpha})$*), the CBECP chooses market risk control. Otherwise, it will carry out credit risk control.*

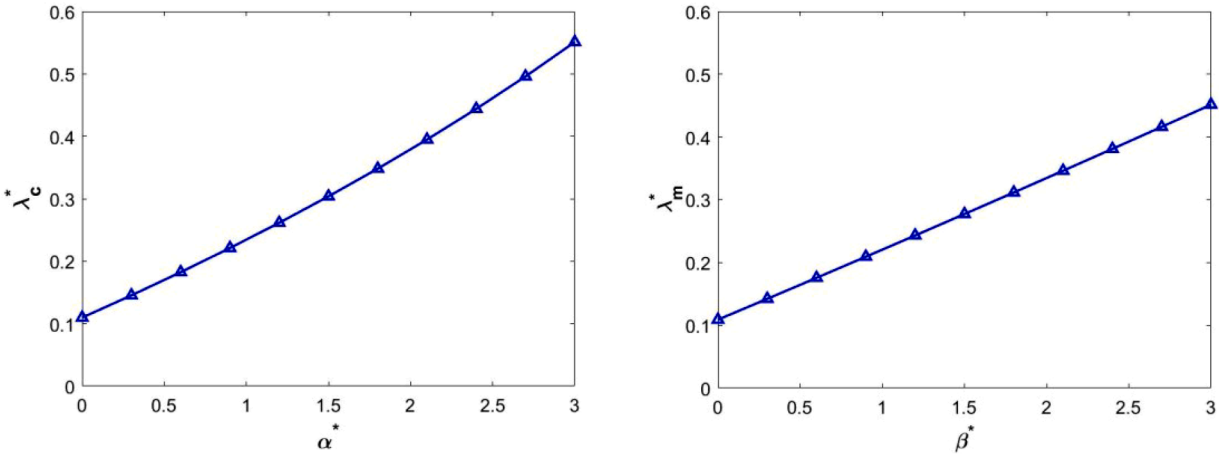
7. Numerical experiments and extensions

7.1. Numerical experiments

We further conduct numerical experiments to verify the key theoretical conclusions obtained. Without loss of generality, we assume that $p = 1$, $Q = 100$, $R = 0.1$, $\xi = 0.8$, $k_c = k_m = \theta = 1$, and the random market demand follows $X \sim \exp(0.01)$. In the "joint pre- and post-loan risk control" scenario, the trends of the optimal PRs of the CBECP regarding the level of credit supervision or product sales assistance are shown in Fig. 2. It can be seen that as the post-loan risk control is strengthened, i.e., α^* or β^* increases, the CBECP will relax its pre-loan risk control, i.e., raising the PR. Thus, in the joint risk control scenario, the CBECP will choose to only strengthen one of the pre-loan and post-loan risk control measures, while relax the other, as suggested in Proposition 4 and Proposition 7(a). Taking the post-loan market risk control case as an example, as shown in Fig. 3 and Fig. 4, when the default probability of the exporter (ξ) or the degree of appreciation of the domestic currency (θ) rises, both the CBECP's optimal PR and the product sales assitance level will decrease, implying that the CBECP will strengthen the pre-loan risk control while weaken the post-loan risk control, which verifies Propositions 7(b) and 7(c), respectively.

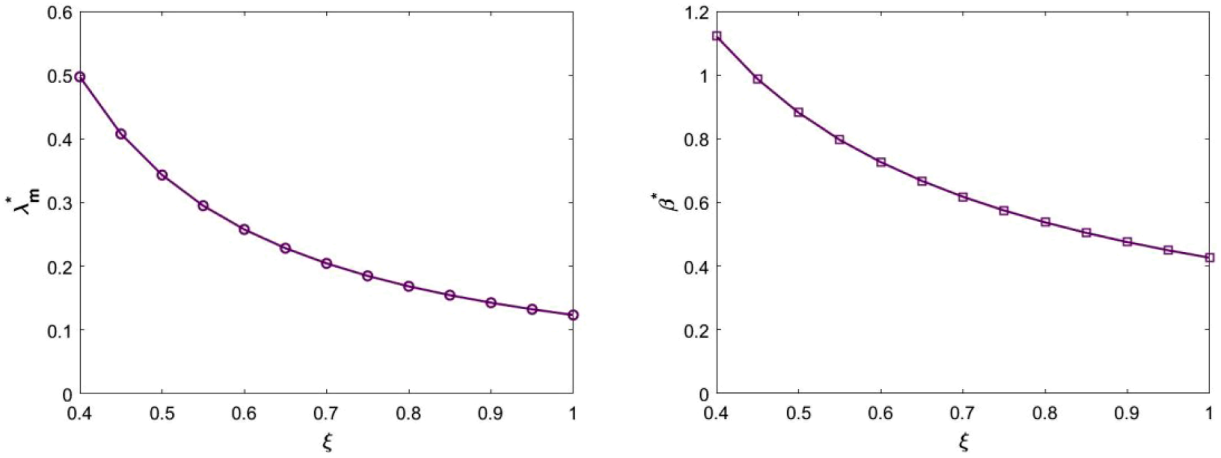
Fig. 5 further shows that under different default probabilities of the exporter, i.e., ξ, the "joint pre- and post-loan risk control" case always brings additional benefits to the CBECP compared with the "only pre-loan risk control" case, whether implementing credit or market risk control strategies in the post-loan stage, which verifies Proposition 8. In addition, from Fig. 5, it can be observed that when ξ is close to 1, the incremental profits brought by the two post-loan risk control measures are approximately the same for the CBECP. However, as ξ becomes smaller, the post-loan credit risk control will become more effective, that is, generating more incremental profits. This means that the CBECP is highly recommended to choose the credit risk control in the post-loan stage when the exporter has better credit.

Fig. 6 further illustrates the choice strategies of the CBECP for post-loan credit or market risk control measures, given it paying the same risk control cost. In such a case, from the expressions of ω in Proposition 9, we have $\omega = 1.041$. Then, we choose $k_c = 0.2$ and $k_m = 0.21$ to meet the case of $k_c \le \omega^2 k_m$, and $\Delta\pi^* = \pi^*_{CRC} - \pi^*_{MRC}$ is observed to be always positive as shown in Fig. 6(a), thus the CBECP always chooses post-loan credit risk control, as suggested by Proposition 9(a). We further choose $k_c = 0.2$ and $k_m = 0.12$ to meet the case of $k_c > \omega^2 k_m$ as shown in Fig. 6(b), it can be observed that there exists a certain threshold $C_{CRC}(\widetilde{\alpha})$, when the CBECP's budget for post-loan risk control is above such a threshold, it still chooses credit risk control, otherwise it chooses market risk control, which fully verifies Proposition 9(b).



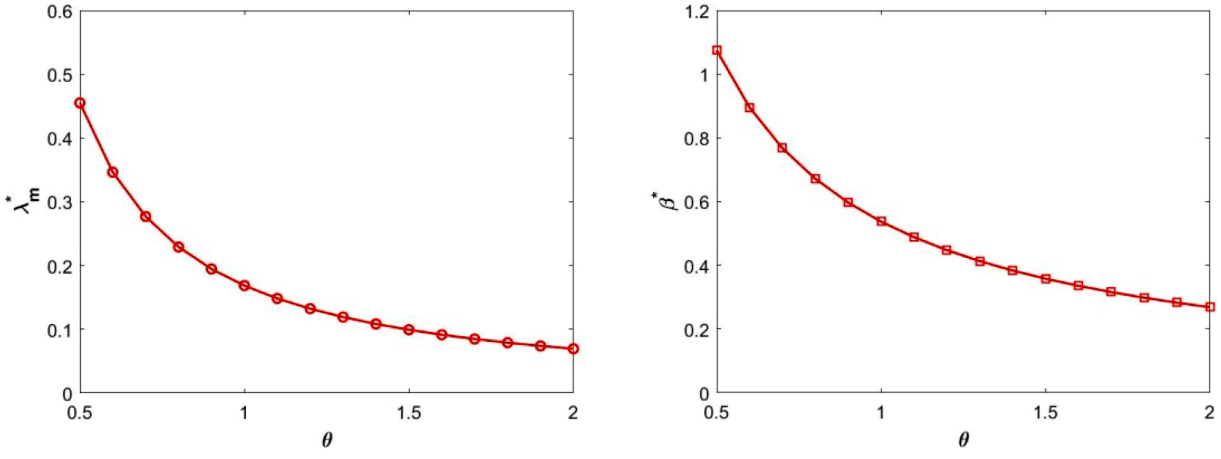
(a) λ_c^* with α^* varying. (b) λ_m^* with β^* varying.

Fig. 2. Relations between PR and credit supervision or product sales assistance level in joint risk control.



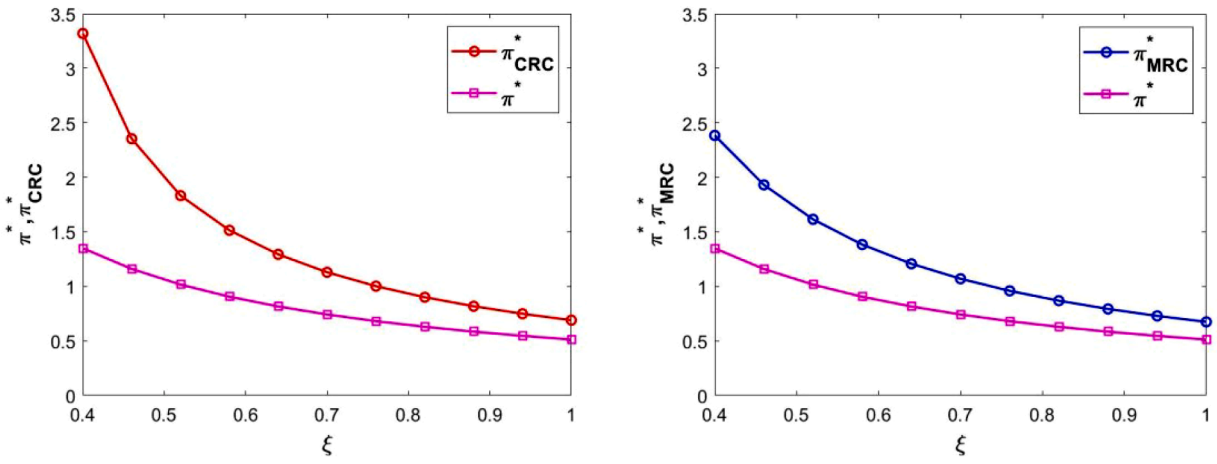
(a) λ_m^* with ξ varying. (b) β^* with ξ varying.

Fig. 3. Relations between exporter's probability of default and PR or product sales assistance level in post-loan credit risk control.



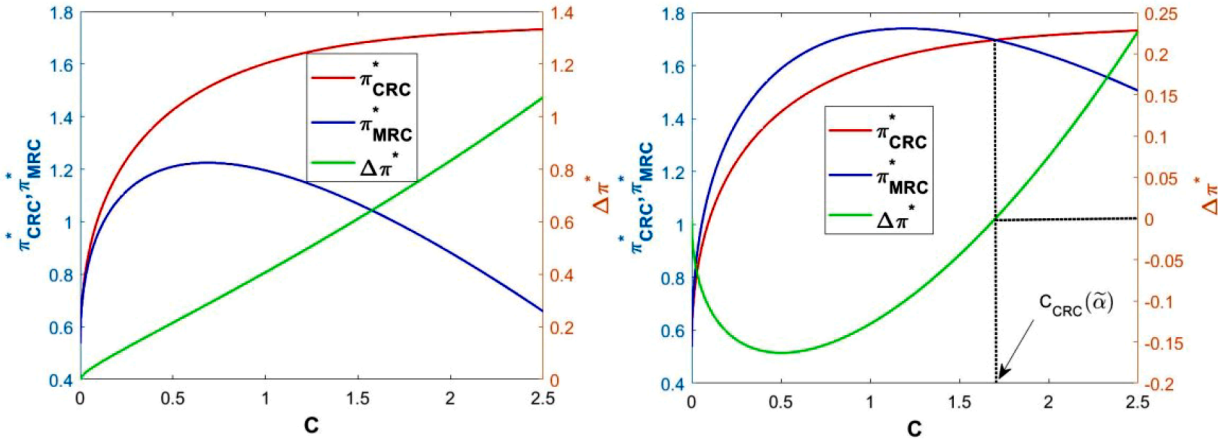
(a) λ_m^* with θ varying. (b) β^* with θ varying.

Fig. 4. Relations between degree of appreciation of the domestic currency and PR or product sales assistance level in post-loan credit risk control.



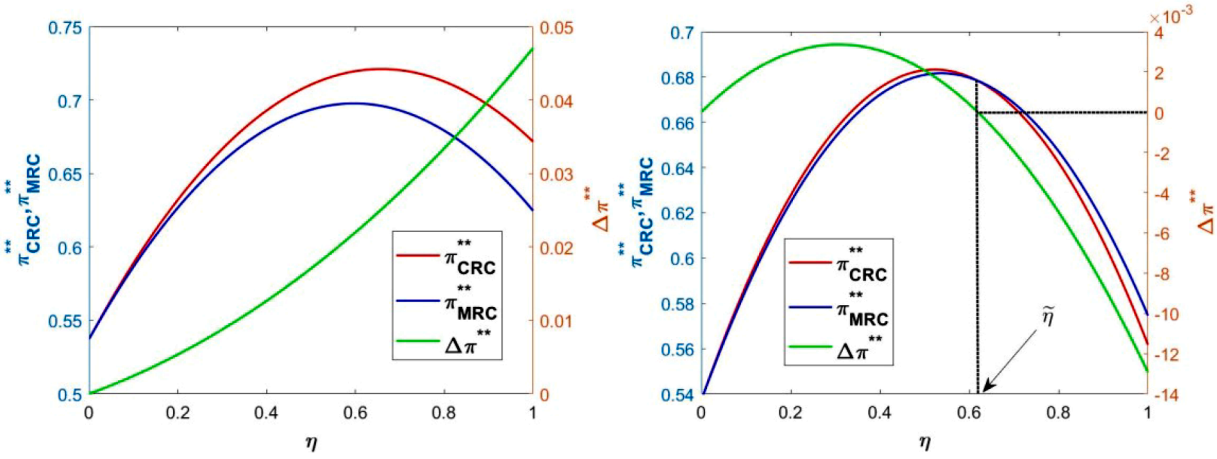
(a) post-loan-credit risk control **(b) post-loan-market risk control**

Fig. 5. Comparisons of profits in "only pre-loan risk control" and "joint pre- and post-loan risk control".



(a) Case of $k_c \le \omega^2 k_m$ **(b) Case of $k_c > \omega^2 k_m$**

Fig. 6. Choice of the CBECP for post-loan credit or market risk control (equal risk control cost case).



(a) Case of $k_c \le k_m + \vartheta$ **(b) Case of $k_c > k_m + \vartheta$**

Fig. 7. Choice of the CBECP for post-loan credit or market risk control (equal risk control level case).

7.2. Extension for the case with equal risk control level

In Section 6, we have answered which post-loan risk control measures are more advantageous for CBECP when it needs to pay *the same risk control cost,* i.e., $C_{CRC}(\alpha) = C_{MRC}(\beta)$. In this section, we further tackle its choice between post-loan credit and market risk control when it needs to pay *the same risk control level,* i.e., $\alpha = \beta$. We assume $\alpha = \beta = \eta$ are exogenously-given. By substituting λ_c^* in Eq. (4) and λ_m^* in Eq. (6) into Eq. (3) and Eq. (5) respectively, we get optimal profits of the CBECP under the two post-loan risk control measures, i.e.,

$$\begin{cases} \pi_{CRC}^{**} = \dfrac{\xi p}{\theta}\left\{\delta F^{-1}[\delta(1+\eta)] - \dfrac{1}{(1+\eta)}\displaystyle\int_0^{F^{-1}[\delta(1+\eta)]} F(x)dx\right\} - \dfrac{1}{2}k_c\eta^2 \\ \pi_{MRC}^{**} = \dfrac{\xi p(1+\eta)}{\theta}\left[\delta F^{-1}(\delta) - \displaystyle\int_0^{F^{-1}(\delta)} F(x)dx\right] - \dfrac{1}{2}k_m\eta^2 \end{cases}, \tag{8}$$

where $\delta = \frac{R}{\xi(1+R)}$. Taking the difference between the two, we have

$$\begin{aligned} &\Delta\pi^{**}(\eta) = \pi_{CRC}^{**} - \pi_{MRC}^{**} \\ &= \frac{\xi p}{\theta}\left\{\delta F^{-1}[\delta(1+\eta)] - \delta F^{-1}(\delta) + (1+\eta)\int_0^{F^{-1}(\delta)} F(x)dx - \frac{1}{(1+\eta)}\int_0^{F^{-1}[\delta(1+\alpha)]} F(x)dx\right\}. \\ &\quad + \frac{1}{2}\eta^2(k_m - k_c) \end{aligned} \tag{9}$$

In this case, different from that in Proposition 9, it becomes difficult to analyze the properties of $\Delta\pi^{**}(\eta)$, thus we cannot theoretically derive the optimal choices for the CBECP towards post-loan credit and market risk control. Thus, we apply numerical experiments to draw observational results by using the same parameter settings in Section 7.1, as shown in the Fig. 7. It can be observed that when the CBECP pays the same level for post-loan credit and market risk control, its optimal choices depend on the cost coefficients of the post-loan credit supervision (k_c) and product sales assistance (k_m), as well as the specific level of the risk control (η). Specifically, when $k_c \le k_m + \vartheta$,[6] $\Delta\pi^* = \pi_{CRC}^* - \pi_{MRC}^*$ is observed to be always positive as shown in Fig. 7(a), thus the CBECP always chooses the post-loan credit risk control. Otherwise, there exists a threshold of the risk control level ($\widetilde{\eta}$) as shown in Fig. 7(b), if the CBECP's post-loan risk control level is below such a threshold, it still chooses credit risk control; otherwise, it chooses market risk control. *In summary, when and only when the market risk control is cheaper enough (cost coefficient of product sales assistance satisfies $k_m < k_c - \vartheta$) and post-loan risk control level is relatively high (risk control level meets $\eta > \widetilde{\eta}$), the CBECP chooses market risk control; otherwise, it will carry out credit risk control.*

Comparing Fig. 7 with Fig. 6, it can be further concluded that when the credit risk control is cheaper enough, i.e., $k_c \le \omega^2 k_m$ or $k_c \le k_m + \vartheta$, the CBECP will always choose credit risk control, regardless of whether it pays the same risk control cost or level. Otherwise, it chooses credit risk control only when the risk control cost is above the threshold $C_{CRC}(\widetilde{\alpha})$ or the risk control level is below the threshold ($\widetilde{\eta}$). Overall, the CBECP will be more likely to apply post-loan credit risk control in the case of a higher risk control cost or a lower risk control level.

[6] Note that ϑ is a constant greater than 0, i.e., $\vartheta > 0$.

8. Conclusions

With the development of big data technology, CBECPs empowered by the digital economy can realize post-loan risk control of exporters in the CBEC-based IPF business. For instance, they can monitor exporters' post-loan business activities on the platform by using big data technology to reduce their "credit default risk", or use search engine optimization, homepage recommendations, and other online marketing technologies to promote the sales of exporters' pledged goods and reduce their "market sales risk". We study the risk control of the CBECP in the CBEC-based IPF business under the empowerment of the digital economy. The pre-loan risk control measures take into account the setting of the PR, while the post-loan risk control measures towards credit and market risks consider, respectively, the platform's online credit supervision and product sales assistance for the exporter. On the basis of considering exchange rate fluctuations, we build decision models under two scenarios, i.e., "only control pre-loan risk" and "joint control pre- and post-loan risk", and find that compared to the former, the latter is more profitable for the CBECP, indicating that the digital economy can empower CBECP to achieve more comprehensive risk control in the CBEC-based IPF business and improve economic benefits. Furthermore, in the case of CBECP paying equal risk control cost, the optimal choices for post-loan credit and market risk control measures are provided. Results show that when and only when the market risk control is cheaper enough and the budget is relatively limited, the CBECP chooses market risk control. Otherwise, it will carry out credit risk control. We also extend the model to study the case when the CBECP pays equal risk control level in the post-loan risk control, and conduct numerical experiments to verify the above conclusions.

These findings suggest CBECPs to adopt joint pre- and post-loan risk control in the CBEC-based IPF business, and provide directions for how CBECPs should choose optimal post-loan risk control measures in the context of the digital economy. The limitation of the paper lies in that we consider the simple operations of the emerging CBEC-based IPF business. Following research can explore more complicated risk control decisions of CBECPs in contexts closer to reality, such as multi-period or multi-product financing, and so on.

CRediT authorship contribution statement

Aolin Leng: Writing – original draft, Supervision, Project administration, Formal analysis, Conceptualization. **Maolin Sun:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Jinzhao Shi:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Acknowledgements

This study was supported by the National Natural Science Foundation of China [grant number 72202021], the Postdoctoral Science

Foundation of China [grant number 2023T160561, 2019M663605], the MOE (Ministry of Education in China) Project of Humanities and Social Sciences for Young Scholars [grant number 20YJC790115], the Innovation Capability Support Plan Project of Shaanxi Province [grant number 2024ZC-YBXM-097], the Fundamental Research Funds for the Central Universities, CHD [grant number 300102233606], and the Shaanxi Provincial Social Science Fund [Research on Shaanxi's Continuous Optimization of Business Environment Empowering High Quality Development Based on Investment Banking Thinking]. We also express our great respect and gratitude to the two anonymous reviewers for their constructive comments, which helped us broaden the scope and enhance the contributions of our paper.

Appendix

Proof of Proposition 1. Calculating the first and second derivatives of π concerning λ, we obtain $\frac{d\pi}{d\lambda} = pQR - \xi pQ(1+R)F(\widehat{x})$, $\frac{d^2\pi}{d\lambda^2} = -\xi pQ^2(1+R)^2\theta f(\widehat{x}) < 0$. Let$\frac{d\pi}{d\lambda} = 0$, we obtain $\lambda^* = \frac{F^{-1}(\delta)}{Q(1+R)\theta}$. □

Proof of Proposition 2. The Hessian matrix of π_{CRC} is

$$H(\lambda_c,\alpha) = \begin{bmatrix} \frac{\partial^2\pi_{CRC}}{\partial\lambda_c{}^2} & \frac{\partial^2\pi_{CRC}}{\partial\lambda_c\partial\alpha} \\ \frac{\partial^2\pi_{CRC}}{\partial\alpha\partial\lambda_c} & \frac{\partial^2\pi_{CRC}}{\partial\alpha^2} \end{bmatrix} = \begin{bmatrix} -\frac{\xi p\widehat{x}_c^2 f(\widehat{x}_c)}{\lambda_c^2\theta(1+\alpha)} & \frac{\xi p\widehat{x}_c F(\widehat{x}_c)}{\lambda_c\theta(1+\alpha)^2} \\ \frac{\xi p\widehat{x}_c F(\widehat{x}_c)}{\lambda_c\theta(1+\alpha)^2} & -\frac{2\xi p\int_0^{\widehat{x}_c}F(x)dx}{\theta(1+\alpha)^3} - k_c \end{bmatrix}. \tag{10}$$

Obvious, the first-order principal minor of $H(\lambda_c,\alpha)$ is negative, while the second-order is

$$|H(\lambda_c,\alpha)| = \frac{\xi^2p^2\widehat{x}_c^2\left[2f(\widehat{x}_c)\int_0^{\widehat{x}_c}F(x)dx - F^2(\widehat{x}_c)\right]}{\lambda_c^2\theta^2(1+\alpha)^4} + \frac{k_c\xi p\widehat{x}_c^2 f(\widehat{x}_c)}{\lambda_c^2\theta(1+\alpha)}. \tag{11}$$

Let $T(\widehat{x}_c) = 2f(\widehat{x}_c)\int_0^{\widehat{x}_c}F(x)dx - F^2(\widehat{x}_c)$, we have $\frac{dT(\widehat{x}_c)}{d\widehat{x}_c} = 2f'(\widehat{x}_c)\int_0^{\widehat{x}_c}F(x)dx$. According to Assumption A5, when $f'(x) = 0$, $\frac{dT(\widehat{x}_c)}{d\widehat{x}_c} = 0$ and $T(\widehat{x}_c) = T(0) = 0$, thus in this case $|H(\lambda_c,\alpha)| > 0$; when $f'(x) < 0$, let $Z(\widehat{x}_c(Q)) = \xi pT(\widehat{x}_c(Q)) + k_c\theta(1+\alpha)^3 f(\widehat{x}_c(Q))$, since $\frac{dZ(\widehat{x}_c)}{d\widehat{x}_c} = \xi p\frac{dT(\widehat{x}_c)}{d\widehat{x}_c} + k_c\theta(1+\alpha)^3 f'(\widehat{x}_c) < 0$ and $\frac{d\widehat{x}_c}{dQ} > 0$, so we have $\frac{dZ(Q)}{dQ} < 0$. Further, based on $Z(Q=0) > 0$ and $Z(Q\to+\infty) < 0$, we infer that there always exists a unique $\widetilde{Q}$ solving $Z(\widetilde{Q}) = 0$, and when $Q < \widetilde{Q}$ is satisfied, $Z(Q) > 0$ always occurs, leading to the truth that $|H(\lambda_c,\alpha)| = \frac{\xi p\widetilde{x}_c^2}{\lambda_c^2\theta^2(1+\alpha)^4}Z(Q) > 0$. It implies that if the pledged level is not extremely larger ($Q < \widetilde{Q}$) or equivalently the loan amount is not extremely high (below $\lambda_c p\widetilde{Q}$), we will have $|H(\lambda_c,\alpha)| > 0$ and $H(\lambda_c,\alpha)$ would be negative definite. From the first derivatives of π_{CRC} concerning λ_c and α, i.e.,

$$\begin{cases} \frac{\partial\pi_{CRC}}{\partial\lambda_c} = pQR - \frac{\xi pQ(1+R)F(\widehat{x}_c)}{1+\alpha} \\ \frac{\partial\pi_{CRC}}{\partial\alpha} = \frac{\xi p\int_0^{\widehat{x}_c}F(x)dx}{\theta(1+\alpha)^2} - k_c\alpha \end{cases}, \tag{12}$$

we can obtain Eq. (4), as the optimum. □

Proof of Proposition 3. From Eq. (4), $\alpha^* > 0$, thus $\lambda_c^* = \frac{F^{-1}[\delta(1+\alpha^*)]}{Q(1+R)\theta} > \frac{F^{-1}[\delta]}{Q(1+R)\theta} = \lambda^*$. □

Proof of Proposition 4. From Eq. (4), we have $\frac{d\lambda_c^*}{d\alpha^*} = \frac{\delta}{Q(1+R)\theta f[\delta(1+\alpha^*)]} > 0$. □

Proof of Proposition 5. The Hessian matrix of π_{MRC} is

$$H(\lambda_m,\beta) = \begin{bmatrix} \frac{\partial^2\pi_{MRC}}{\partial\lambda_m{}^2} & \frac{\partial^2\pi_{MRC}}{\partial\lambda_m\partial\beta} \\ \frac{\partial^2\pi_{MRC}}{\partial\beta\partial\lambda_m} & \frac{\partial^2\pi_{MRC}}{\partial\beta^2} \end{bmatrix} = \begin{bmatrix} -\frac{\xi p\widehat{x}_m^2(1+\beta)f(\widehat{x}_m)}{\lambda_m^2\theta} & \frac{\xi p\widehat{x}_m^2 f(\widehat{x}_m)}{\lambda_m\theta} \\ \frac{\xi p\widehat{x}_m^2 f(\widehat{x}_m)}{\lambda_m\theta} & -\frac{\xi p\widehat{x}_m^2 f(\widehat{x}_m)}{\theta(1+\beta)} - k_m \end{bmatrix}. \tag{13}$$

Obvious, the first-order principal minor of $H(\lambda_m,\beta)$ is negative, while the second-order is $|H(\lambda_m,\beta)| = \frac{k_m\xi p\widehat{x}_m^2(1+\beta)f(\widehat{x}_m)}{\lambda_m^2\theta} > 0$. Thus, $H(\lambda_c,\alpha)$ is negative definite, and we can get the optimal solution from the first-order conditions, i.e.,

$$\begin{cases} \frac{\partial\pi_{MRC}}{\partial\lambda_m} = pQR - \xi pQ(1+R)F(\widehat{x}_m) \\ \frac{\partial\pi_{MRC}}{\partial\beta} = \frac{\xi p\left[\widehat{x}_m F(\widehat{x}_m) - \int_0^{\widehat{x}_m}F(x)dx\right]}{\theta} - k_m\beta \end{cases}, \tag{14}$$

as shown in Eq. (6). □

Proof of Proposition 6. From Eq. (6), since for any $x < F^{-1}(\delta)$, $F(x) < \delta$ establishes, thus we have $\int_0^{F^{-1}(\delta)}F(x)dx < \delta F^{-1}(\delta)$, and equivalently $\beta^* > 0$. Then, we obtain $\lambda_m^* = \frac{F^{-1}(\delta)(1+\beta^*)}{Q(1+R)\theta} > \frac{F^{-1}(\delta)}{Q(1+R)\theta} = \lambda^*$. □

Proof of Proposition 7. **For part (a).** From Eq. (6), we have $\frac{d\lambda_m^*}{d\beta^*} = \frac{F^{-1}(\delta)}{Q(1+R)\theta} > 0$.

For part (b). From Eq. (6), we have $\frac{d\beta^*}{d\xi} = \frac{\partial\beta^*}{\partial\xi} + \frac{\partial\beta^*}{\partial\delta}\frac{d\delta}{d\xi} = -\frac{p\int_0^{F^{-1}(\delta)}F(x)dx}{\theta k_m} < 0$, and $\frac{d\lambda_m^*}{d\xi} = \frac{\partial\lambda_m^*}{\partial\delta}\frac{d\delta}{d\xi} + \frac{\partial\lambda_m^*}{\partial\beta^*}\frac{d\beta^*}{d\xi} = \underbrace{-\frac{\delta(1+\beta^*)}{\xi Q(1+R)\theta f(\delta)}}^{<0} + \underbrace{\frac{\partial\lambda_m^*}{\partial\beta^*}}^{>0}\underbrace{\frac{d\beta^*}{d\xi}}^{<0} < 0$.

For part (c). From Eq. (6), $\frac{d\beta^*}{d\theta} < 0$, and $\frac{d\lambda_m^*}{d\theta} = \underbrace{\frac{\partial\lambda_m^*}{\partial\theta}}^{<0} + \underbrace{\frac{\partial\lambda_m^*}{\partial\beta^*}}^{>0}\underbrace{\frac{d\beta^*}{d\theta}}^{<0} < 0$. □

Proof of Proposition 8. For the "post-loan credit risk control" case, from Eq. (3), we have

$$\frac{d\pi_{CRC}(\lambda_c=\lambda^*)}{d\alpha} = \frac{\xi p}{\theta(1+\alpha)^2}\int_0^{F^{-1}(\delta)}F(x)dx - k_c\alpha. \tag{15}$$

Since $Lim_{\alpha\to 0}\frac{d\pi_{CRC}(\lambda_c=\lambda^*)}{d\alpha} = \frac{\xi p}{\theta}\int_0^{F^{-1}(\delta)}F(x)dx > 0$ and α^* in Eq. (4) is the optimum, we can infer that as α increases from 0 to α^*, the value of π_{CRC} gradually rises. So, compared with the "only pre-loan risk control" in Section 4 where α can be viewed as 0, the optimal profit of the CBECP in the "joint pre-loan and post-loan-credit risk control" will undoubtedly increase.

For the "post-loan market risk control" case, substituting Eq. (6) into Eq. (5) and Eq. (2) into Eq. (1), we can respectively obtain $\pi_{MRC}^* = \frac{\xi p(1+\beta^*)}{\theta}\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right] - \frac{1}{2}k_m(\beta^*)^2$ and $\pi^* = \frac{\xi p}{\theta}\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]$. Then, we have

$$\begin{aligned}
&\pi_{MRC}^* - \pi^* \\
&= \frac{\xi p\beta^*}{\theta}\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right] - \frac{1}{2}k_m(\beta^*)^2 \\
&= \frac{\xi p\beta^*}{\theta}\frac{\theta k_m\beta^*}{\xi p} - \frac{1}{2}k_m(\beta^*)^2 \\
&= \frac{1}{2}k_m(\beta^*)^2 > 0
\end{aligned}, \tag{16}$$

which shows the joint risk control brings additional benefits to the CBECP, in the "joint pre-loan and post-loan-market risk control" scenario. □

Proof of Proposition 9. From $C_{CRC}(\alpha) = C_{MRC}(\beta) = C$, we have $\frac{1}{2}k_c(\alpha)^2 = \frac{1}{2}k_m(\beta)^2$ and $\beta = \sqrt{\frac{k_c}{k_m}}\alpha$. Let $\Omega = \sqrt{\frac{k_c}{k_m}}$, we have $\beta = \Omega\alpha$. In such a case, α and β are all exogenous, by substituting λ_c^* in Eq. (4) and λ_m^* in Eq. (6) into Eq. (3) and Eq. (5) respectively, we get optimal profits of the CBECP under post-loan credit and market risk control as

$$\begin{cases}
\pi_{CRC}^* = \frac{\xi p}{\theta}\left\{\delta F^{-1}[\delta(1+\alpha)] - \frac{1}{(1+\alpha)}\int_0^{F^{-1}[\delta(1+\alpha)]}F(x)dx\right\} - \frac{1}{2}k_c(\alpha)^2 \\
\pi_{MRC}^* = \frac{\xi p(1+\Omega\alpha)}{\theta}\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right] - \frac{1}{2}k_c(\alpha)^2
\end{cases}. \tag{17}$$

Taking the difference between the two, we have

$$\Delta\pi^*(\alpha) = \pi_{CRC}^* - \pi_{MRC}^* = \frac{\xi p}{\theta}\left\{\delta F^{-1}[\delta(1+\alpha)] - \frac{1}{(1+\alpha)}\int_0^{F^{-1}[\delta(1+\alpha)]}F(x)dx\right\} - \frac{\xi p(1+\Omega\alpha)}{\theta}\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]. \tag{18}$$

Taking the first and second derivatives of $\Delta\pi^*(\alpha)$ concerning α, we have

$$\begin{cases}
\frac{d\Delta\pi^*(\alpha)}{d\alpha} = \frac{\xi p}{\theta}\left\{\frac{\int_0^{F^{-1}[\delta(1+\alpha)]}F(x)dx}{(1+\alpha)^2} - \Omega\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]\right\} \\
\frac{d^2\Delta\pi^*(\alpha)}{d\alpha^2} = \frac{\xi p\left\{\frac{\delta^2(1+\alpha)^2}{f[\delta(1+\alpha)]} - 2\int_0^{F^{-1}[\delta(1+\alpha)]}F(x)dx\right\}}{\theta(1+\alpha)^3}
\end{cases}. \tag{19}$$

Next, we prove that $\Delta\pi^*(\alpha)$ is convex in α over $\alpha\in[0,+\infty)$, that is, $\frac{d^2\Delta\pi^*(\alpha)}{d\alpha^2} \geq 0$. First, let $\mu = \delta(1+\alpha)$, then $N(\mu) = \frac{\delta^2(1+\alpha)^2}{f[\delta(1+\alpha)]} - 2\int_0^{F^{-1}[\delta(1+\alpha)]}F(x)dx = \frac{\mu^2}{f(\mu)} - 2\int_0^{F^{-1}(\mu)}F(x)dx$. Since $\frac{dN(\mu)}{d\mu} = -\frac{\mu^2 f'(\mu)}{[f(\mu)]^2} \geq 0$ under the Assumption A5, in conjunction with $N(\mu=0) = 0$, so we conclude that $N(\mu)\geq 0$ over $\mu\in[0,+\infty)$. Since $\delta = \frac{R}{\xi(1+R)} > 0$, $\mu = \delta(1+\alpha) > 0$ establishes over $\alpha\in[0,+\infty)$. Thus for $\alpha\in[0,+\infty)$, $N(\mu)\geq 0$ is true, which yields $\frac{d^2\Delta\pi^*(\alpha)}{d\alpha^2} \geq 0$.

Further, let $\omega = \frac{\int_0^{F^{-1}(\delta)}F(x)dx}{\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx}$. From Assumption A5, the curve of $F(x)$ is concave, so we have $\int_0^{F^{-1}(\delta)}F(x)dx \geq \frac{1}{2}\delta F^{-1}(\delta)$. Thus, $\omega \geq 1$ always establishes.

For part (a). If $k_c \leq \omega^2 k_m$, $\Omega = \sqrt{\frac{k_c}{k_m}} \leq \omega$ holds. Then from Eq. (19), we have

$$\begin{aligned}
&\lim_{\alpha\to 0}\frac{d\Delta\pi^*(\alpha)}{d\alpha} = \frac{\xi p}{\theta}\left\{\int_0^{F^{-1}(\delta)}F(x)dx - \Omega\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]\right\} \\
&\geq \frac{\xi p}{\theta}\left\{\int_0^{F^{-1}(\delta)}F(x)dx - \omega\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)}F(x)dx\right]\right\} = 0
\end{aligned}. \tag{20}$$

Recall the convexity of $\Delta\pi^*(\alpha)$ over $\alpha \in [0, +\infty)$ as proved, and the truth $\Delta\pi^*(\alpha = 0) = 0$, we conclude that in this case $\Delta\pi^*(\alpha) \geq 0$ always true over $\alpha \in [0, +\infty)$, that is, $\pi^*_{CRC} \geq \pi^*_{MRC}$ and the CBECP chooses post-loan credit risk control.

For part (b). If $k_c > \omega^2 k_m$, $\Omega = \sqrt{\frac{k_c}{k_m}} > \omega$ holds. Then from Eq. (19), we have

$$\begin{aligned} \lim_{\alpha \to 0} \frac{d\Delta\pi^*(\alpha)}{d\alpha} &= \frac{\xi p}{\theta}\left\{\int_0^{F^{-1}(\delta)} F(x)dx - \Omega\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)} F(x)dx\right]\right\} \\ &< \frac{\xi p}{\theta}\left\{\int_0^{F^{-1}(\delta)} F(x)dx - \omega\left[\delta F^{-1}(\delta) - \int_0^{F^{-1}(\delta)} F(x)dx\right]\right\} = 0 \end{aligned} \tag{21}$$

Further, when $\alpha \to \frac{1-\delta}{\delta}$, $\frac{d\Delta\pi^*(\alpha)}{d\alpha} \to +\infty$. Combining the truth that $\Delta\pi^*(\alpha = 0) = 0$, we can infer that there exists a certain threshold $\widetilde{\alpha} \in \left(0, \frac{1-\delta}{\delta}\right)$ that solves $\Delta\pi^*(\widetilde{\alpha}) = 0$, and when $\alpha < \widetilde{\alpha}$ (equivalent to $C < C_{CRC}(\widetilde{\alpha})$), $\Delta\pi^*(\alpha) < 0$ and $\pi^*_{CRC} < \pi^*_{MRC}$ establishes, the CBECP chooses post-loan market risk control; when $\alpha > \widetilde{\alpha}$ (equivalent to $C > C_{CRC}(\widetilde{\alpha})$), $\Delta\pi^*(\alpha) > 0$ and $\pi^*_{CRC} > \pi^*_{MRC}$ establishes, the CBECP chooses post-loan credit risk control. □

Data availability

No data was used for the research described in the article.

References

[1] Wang X, Xie J, Fan ZP. B2C cross-border e-commerce logistics mode selection considering product returns. Int. J. Prod. Res. 2021;59(13):3841–60.
[2] Wang Y, Jia F, Schoenherr T, et al. Cross-border e-commerce firms as supply chain integrators: the management of three flows. Ind. Mark. Manage. 2020;89:72–88.
[3] Wang F, Yang X, Zhuo X, et al. Joint logistics and financial services by a 3PL firm: effects of risk preference and demand volatility. Transp. Res. Part E 2019;130:312–28.
[4] Yu H, Zhao Y, Liu Z, et al. Research on the financing income of supply chains based on an e-commerce platform. Technol. Forecast Soc. Change 2021;169:120820.
[5] Zhi B, Wang X, Xu F. Impawn rate optimisation in inventory financing: a canonical vine copula-based approach. Int. J. Prod. Econ. 2020;227:107659.
[6] He J, Jiang X, Wang J, et al. VaR methods for the dynamic impawn rate of steel in inventory financing under autocorrelative return. Eur. J. Oper. Res. 2012;223(1):106–15.
[7] Zhi B, Wang X, Xu F. Managing inventory financing in a volatile market: a novel data-driven copula model. Transp. Res. Part E 2022;165:102854.
[8] Sim J, Prabhu V. A microcredit contract model with a Black–Scholes model under default risk. Int. J. Prod. Econ. 2017;193:294–305.
[9] Zhang W, Yan S, Li J, et al. Credit risk prediction of SMEs in supply chain finance by fusing demographic and behavioral data. Transp. Res. Part E 2022;158:102611.
[10] Buzacott JA, Zhang RQ. Inventory management with asset-based financing. Manage. Sci. 2004;50(9):1274–92.
[11] Li L, Shubik M, Sobel MJ. Control of dividends, capital subscriptions, and physical inventories. Manage Sci. 2013;59(5):1107–24.
[12] Chang S, Li A, Wang X, et al. Joint optimization of e-commerce supply chain financing strategy and channel contract. Eur. J. Oper. Res. 2022;303(2):908–27.
[13] Sokolinskiy O, Melamed B, Sopranzetti B. Precautionary replenishment in financially constrained inventory systems subject to credit rollover risk and supply disruption. Ann. Oper. Res. 2018;271(2):971–97.
[14] Xie X, Shi X, Gu J, et al. Examining the contagion effect of credit risk in a supply chain under trade credit and bank loan offering. Omega (Westport) 2023;115:102751.
[15] Yang Y, Bi G, Liu L. Profit allocation in investment-based crowdfunding with investors of dynamic entry times. Eur. J. Oper. Res. 2020;280(1):323–37.
[16] Deng S, Fu K, Xu J, et al. The supply chain effects of trade credit under uncertain demands. Omega (Westport) 2021;98:102113.
[17] Kouvelis P, Zhao W. Financing the newsvendor: supplier vs. bank, and the structure of optimal trade credit contracts. Oper. Res. 2012;60(3):566–80.
[18] Wang W, Cai G. Curtailing bank loan and loan insurance under risk regulations in supply chain finance. Manage. Sci. 2024;70(4):2682–98.
[19] Kouvelis P, Li R. Integrated risk management for newsvendors with value-at-risk constraints. Manuf. Serv. Oper. Manage. 2019;21(4):816–32.
[20] Cao E, Yu M. The bright side of carbon emission permits on supply chain financing and performance. Omega (Westport) 2019;88:24–39.
[21] Dong L, Qiu Y, Xu F. Blockchain-enabled deep-tier supply chain finance. Manuf. Serv. Oper. Manage. 2023;25(6):2021–37.
[22] Huang X, Boyacı T, Gümüş M, et al. United we stand or divided we stand? Strategic supplier alliances under order default risk. Manage Sci. 2016;62(5):1297–315.
[23] Li J, Zhang S, Hou P. The value of backup supply for mitigating disruption risk with equity finance. Int. J. Prod. Res. 2024:1–28 [Early Access].
[24] Zha Y, Chen K, Dong L, et al. Financing supplier through retailer's credit, 33. Production and Operations Management; 2024. p. 721–36.
[25] Xiao S, Sethi SP, Li G. Trade credit vs. supplier-guaranteed financing: role of friction costs. Transp. Res. Part E 2024;183:103421.
[26] Zhong H, Yan N, Lu J, et al. Value of loan credit insurance in the capital-constrained supply chain. Ann. Oper. Res. 2024 [Early Access].
[27] Yu H, Wang Q. Supply chain analysis about enterprise's exchange rate risk management based on the risk-taking mechanism. Chin. J. Manage. Sci. 2021;29(09):44–53 (in Chinese).
[28] Shi J, Liu D, Du Q, et al. The role of the procurement commitment contract in a low-carbon supply chain with a capital-constrained supplier. Int. J. Prod. Econ. 2023;255:108681.
[29] Chen X. A model of trade credit in a capital-constrained distribution channel. Int. J. Prod. Econ. 2015;159:347–57.
[30] Shi J, Du Q, Lin F, et al. Coordinating the supply chain finance system with buyback contract: a capital-constrained newsvendor problem. Comput. Ind. Eng. 2020;146:106587.
[31] Shi J, Du Q, Lin F, et al. Optimal financing mode selection for a capital-constrained retailer under an implicit bankruptcy cost. Int. J. Prod. Econ. 2020;228:107657.